EDS 223 Homework Assignment 2

Exploring patterns of environmental justice

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Load Packages and Source Functions

```
library(tidyverse)
library(sf)
library(stars)
library(tmap)
library(here)
library(gt)
```

Import Data

```
# Read in HOLC grade designations for Los Angeles
redline_LA <- sf::st_read(here::here("data", "mapping-inequality", "mapping-inequality-los-a
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/home/rachelcmiller/EDS223-HW/EDS223-HW2/data/mapping-inequality/mapping-
inequality-los-angeles.json'
```

```
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```
# Read in ejscreen data
ej_screen <- sf::st_read(here::here("data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CI
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/home/rachelcmiller/EDS223-HW/EDS223-HW2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_C
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# Read in crash data
crashes <- read_csv(here::here("data", "SWITRS_LA", "Crashes.csv"))
```

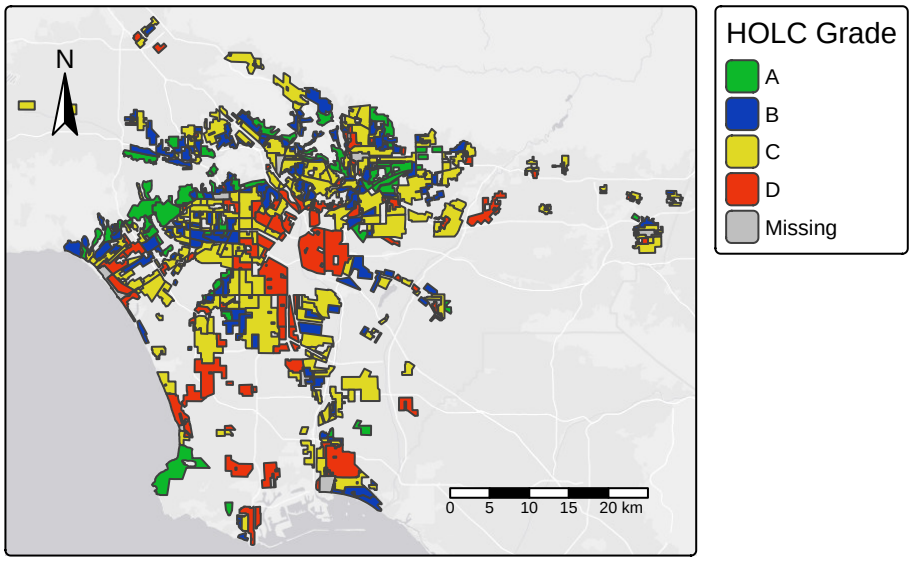
```
# Filter to Los Angeles
los_angeles <- ej_screen |>
  dplyr::filter(CNTY_NAME == 'Los Angeles County')
```

Part 1: Legacy of redlining in current environmental (in)justice

```
# Create a map of historical redlining neighborhoods
grade_colors <- c("#0DB82A", "#0D3DB8","#E0D924","#EB340E")

tm_shape(redline_LA) +
   tm_polygons(
    fill = 'grade',
    fill.scale = tm_scale(values = grade_colors),
    fill.legend = tm_legend(title = "HOLC Grade")) +
    tm_basemap() +
    tm_title("Fig.1 : Historical HOLC Grade Rankings in Los Angeles") +
  tm_scalebar() +
  tm_compass(position = c("left", "top"))
```

Fig.1 : Historical HOLC Grade Rankings in Los Angeles




```
# Match crs of redline_LA and los_angeles
st_crs(redline_LA) == st_crs(los_angeles)
```

```
[1] FALSE
```

```
redline_LA_transform <- st_transform(redline_LA, crs = st_crs(los_angeles))

# Verify that crs match
if(st_crs(redline_LA) == st_crs(los_angeles)){
  print("CRS MATCH")
}else{
  warning("CRS NOT A MATCH")
}

# Combine datasets
redline_joined <- st_join(redline_LA_transform, los_angeles)
```

```
# Summarise percentage of census block groups in each HOLC grade
percent_HOLC <- redline_joined |> st_drop_geometry() |>
  group_by(grade) |>
  summarise(percentage = n()/nrow(redline_joined) * 100)
```

Percentage of Census Block Groups in each HOLC Grade

grade	percentage
A	7.028804
B	19.395742
C	47.871008
D	21.070758
NA	4.633688

```
percent_HOLC |>
  gt() |>
    tab_header(title = md("Percentage of Census Block Groups in each HOLC Grade"))
```

```
# Check for NAs
sum(is.na(redline_joined$P_LOWINCPCT))
```

```
[1] 0
```

```
sum(is.na(redline_joined$P_PM25))
```

```
[1] 0
```

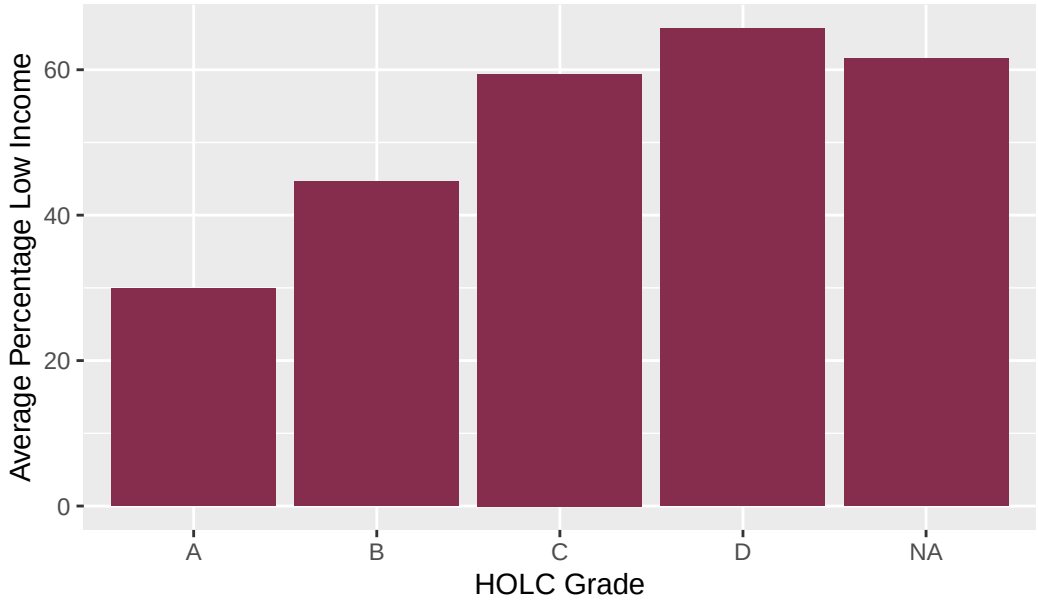
```
sum(is.na(redline_joined$P_LIFEEXPPCT))
```

```
[1] 294
```

```
# Calculate mean percentage low income population in each HOLC grade
mean_lowinc <- redline_joined |>
  group_by(grade) |>
  summarise(mean_low_inc_PCT = mean(P_LOWINCPCT)) |>
  st_drop_geometry()

# Visualize the mean percentage low income population per HOLC grade
mean_lowinc |>
  ggplot(aes(y=mean_low_inc_PCT, x=grade)) +
  geom_bar(stat="identity", fill = "#862C4D") +
  labs(title = "Fig.2 : Average Percent Low Income Population Per HOLC Grade", x = "HOLC Grad
```

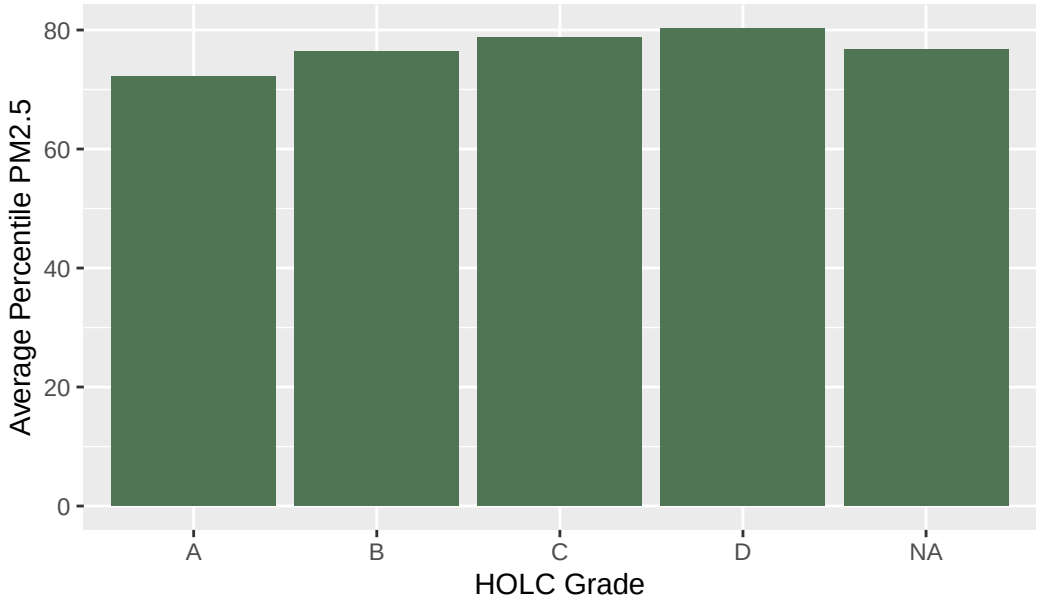
Fig.2 : Average Percent Low Income Population Per HOLC Gra




```
# Calculate mean percentile for Particulate Matter 2.5 for each HOLC grade
mean_pm25 <- redline_joined |>
  group_by(grade) |>
  summarise(mean_pm_25 = mean(P_PM25)) |>
  st_drop_geometry()

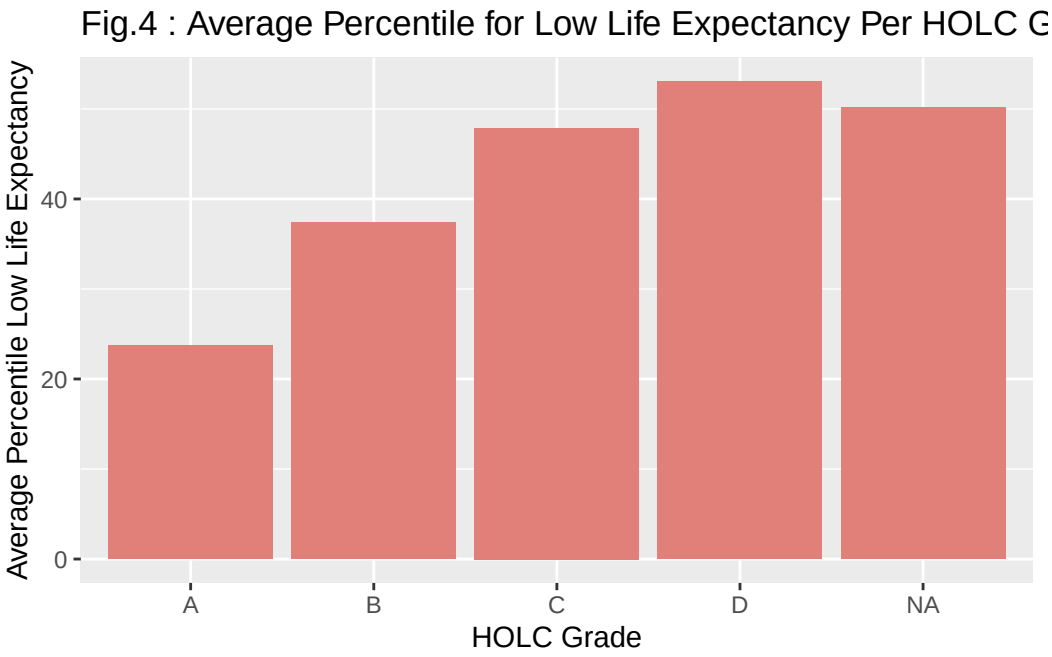
# Visualize the mean percentile for Particulate Matter 2.5 for each HOLC grade
mean_pm25 |>
  ggplot(aes(y=mean_pm_25, x=grade)) +
  geom_bar(stat="identity", fill = "#507555") +
  labs(title = "Fig.3: Average Percentile for Particulate Matter 2.5 Per HOLC Grade", x = "HO
```

Fig.3: Average Percentile for Particulate Matter 2.5 Per HOLC G




```
# Calculate mean percentile for low life expectancy for each HOLC grade
mean_life_exp <- redline_joined |>
  group_by(grade) |>
  summarise(mean_life_exp_pct = mean(P_LIFEEXPPCT, na.rm=TRUE)) |> # drop NAs
  st_drop_geometry()

# Visualize the mean percentile for low life expectancy for each HOLC grade
mean_life_exp |>
  ggplot(aes(y=mean_life_exp_pct, x=grade)) +
  geom_bar(stat="identity", fill = "#E18079") +
  labs(title = "Fig.4 : Average Percentile for Low Life Expectancy Per HOLC Grade", x = "HOLC
```



Reflection on Results

Figure 1 shows the distribution of Home Owner's Loan Corporation (HOLC) grades across the Los Angeles area. In the 1930s , the HOLC graded neighborhoods as either A (green), B (blue), C (yellow) or D (red) corresponding to the perceived risk associated with insuring mortgages, known as "redlining". Figure 2 represents the mean percentage low income population in each grade in Los Angeles. From this figure, a less favorable HOLC grade is associated with a greater percentage of low income. Figures 3 and 4 display the average percentile per HOLC grade for particulate matter 2.5 and low life expectancy respectively. There is a marked increase in both percentiles as the grade moves from A to D, suggesting a correlation between a less favorable HOLC grade and less favorable environmental and health outcomes. The increase in average percentile for low life expectancy between grades is more sizable than the increase in average percentile for particulate matter 2.5. This may suggest that other factors than particulate matter 2.5 are also impacting the life expectancy. Overall, Figures 1 to 4 demonstrate some of the lasting impacts of the historical practice of "redlining".

Part 2: Legacy of redlining in pedestrian and bike accidents

```
# Check for NAs in POINT_X/POINT_Y and LONGITUDE/LATITUDE
sum(crashes$POINT_X == 'NA')
```

```
[1] NA
```

```
sum(crashes$POINT_Y == 'NA')
```

```
[1] NA
```

```
sum(crashes$LONGITUDE == 'NA')
```

```
[1] NA
```

```
sum(crashes$LATITUDE == 'NA')
```

```
[1] NA
```

```
# Filter crash data to pedestrian or bicycle accidents and filter out NAs
crashes_filtered <- crashes |>
  filter(PEDESTRIAN_ACCIDENT == "Y" | BICYCLE_ACCIDENT == "Y") |>
  filter(POINT_X != 'NA')

# Convert X and Y columns to points using WGS 84
crashes_points <- st_as_sf(crashes_filtered, coords = c("POINT_X", "POINT_Y"), crs = 4326)

# Check if crs match
st_crs(redline_joined) == st_crs(crashes_points)
```

```
[1] FALSE
```

```
# Match crs and verify match
crashes_transform <- st_transform(crashes_points, crs = st_crs(redline_joined))
if(st_crs(crashes_transform) != st_crs(redline_joined)) stop('Check if CRS match')

# Subset to points in HOLC regions
crashes_holc <- crashes_transform[redline_joined,]

# Join data frames by left join
crashes_lj <- st_join(crashes_holc, redline_joined) |>
  select(id = ID, region = grade)

# Calculate number of crashes per HOLC grade region
```

```r
crashes_per_grade <- crashes_lj |>
  filter(!is.na(region)) |>
  group_by(region) |>
  summarise(number_points = n()) |>
  st_drop_geometry()

# Calculate percentage crashes in each grade region
crashes_percent_grade <- crashes_per_grade |>
  mutate(Percentage = 100*(number_points/sum(crashes_per_grade$number_points)))

# Convert region column to factors
crashes_percent_grade$region <- as.factor(crashes_percent_grade$region)

# Visualize the percentage crashes per HOLC region
crashes_percent_grade |>
  ggplot(aes(x = region, y = Percentage, fill = region))+
  geom_bar(stat = "identity", fill = "#9EC0EB") +
  labs(title = "Percentage Bike and Pedestrian Crashes and Fatalities Within Each HOLC Grade
```

